

Education

Ph.D., Princeton University. Astrophysical Sciences. 2022 - 2027
H.B.Sc., University of Toronto. Astronomy & Astrophysics and Statistics. 2018 - 2022

Publications

-
7. **Jeff Shen**, Joshua S. Speagle, Neige Frankel, et. al. *Stellar parameters for 220m stars from Gaia DR3 BP/RP spectra*, in prep.
 6. Yan Liang, Peter Melchior, ChangHoon Hanh, **Jeff Shen**, Andy Goulding, Charlotte Ward. *Outlier Detection in the DESI Bright Galaxy Survey*, submitted to *ApJL*. arXiv: [2307.07664](https://arxiv.org/abs/2307.07664).
 5. **Jeff Shen**, Joshua S. Speagle, Neige Frankel, J. Ted Mackereth, Yuan-Sen Ting, Jo Bovy. *Disentangling Stellar Age Estimates from Galactic Chemodynamical Evolution*, submitted to *ApJ*. arXiv: [2305.15634](https://arxiv.org/abs/2305.15634).
 4. Seery Chen, Deborah M. Lokhorst, **Jeff Shen**, Imad Pasha, Evgeni I. Malakhov, Roberto Abraham, Pieter van Dokkum. *The Dragonfly Spectral Line Mapper: design and first light*. 2022, Proc. SPIE 12182, Ground-based and Airborne Telescopes IX, 121824E. arXiv: [2209.07489](https://arxiv.org/abs/2209.07489).
 3. Deborah M. Lokhorst, Seery Chen, Imad Pasha, **Jeff Shen**, Evgeni I. Malakhov, Roberto G. Abraham, Pieter van Dokkum. *The pathfinder Dragonfly Spectral Line Mapper: pushing the limits for ultra-low surface brightness spectroscopy*. 2022, Proc. SPIE 12182, Ground-based and Airborne Telescopes IX, 121821T. arXiv: [2209.07487](https://arxiv.org/abs/2209.07487)
 2. **Jeff Shen**, Gwendolyn M. Eadie, Norman Murray, Dennis Zaritsky, Joshua S. Speagle, Yuan-Sen Ting, Charlie Conroy, Phillip A. Cargile, Benjamin D. Johnson, Rohan P. Naidu, Jiwon Jesse Han. *The Mass of the Milky Way from the H3 Survey*. 2022, *ApJ*, 925, 1. arXiv: [2111.09327](https://arxiv.org/abs/2111.09327).
 1. **Jeff Shen**, Allison W. S. Man, Johannes Zabl, Zhi-Yu Zhang, Mikkel Stockmann, Gabriel Brammer, Katherine E. Whitaker, and Johan Richard. *Molecular gas in a gravitationally lensed galaxy group at $z = 2.9$* . 2021, *ApJ*, 917, 79. arXiv: [2105.11572](https://arxiv.org/abs/2105.11572).

Honours and Awards

2022 Jan	Astrostatistics Student Paper Competition Finalist Astrostatistics Interest Group of the American Statistical Association
2021 - 2022	Dean's List Scholar University of Toronto
2021 May - 2021 Aug	Undergraduate Summer Research Fellowship, (\$8,396 CAD) Canadian Institute for Theoretical Astrophysics, University of Toronto
2020 May - 2020 Aug	Summer Undergraduate Research Program Award (\$9,500 CAD) Dunlap Institute for Astronomy and Astrophysics, University of Toronto

Talks

- 2022 August **JSM 2022, Washington, D.C.**
Contributed talk: “The Mass of the Milky Way from the H3 Survey.”
- 2021 August **SDSS 2021 Collaboration Meeting, JHU/Online**
Lightning talk: “Predicting the ages of stars with machine learning.”
- 2021 May **Stellar Stats Workshop, UofT/Online**
Lightning talk: “Estimating the mass distribution of the Milky Way with Bayesian multilevel models.”
- 2021 May **Canadian Astronomical Society (CASCA) AGM, NRC Herzberg/Online**
Lightning talk: “Molecular gas in a gravitationally lensed galaxy group at $z = 2.9$.”
- 2021 Apr **Vancouver Area Cosmology Meeting, UBC/SFU/TRIUMF/Online**
“Molecular gas in gravitationally lensed galaxies at $z = 2.9$.”
- 2020 Oct **REQUIEM Galaxies Telecon, Online**
“Molecular gas in gravitationally lensed galaxies at $z = 2.9$.”

Posters

- 2021 May **Canadian Astronomical Society (CASCA) AGM, NRC Herzberg/Online**
“Molecular gas in a gravitationally lensed galaxy group at $z = 2.9$.”
- 2020 Aug **Summer Undergraduate Research Program Poster Session, UofT/Online**
“Molecular gas in a gravitationally lensed galaxy group at $z = 2.9$.” Awarded prize for one of top three posters.

Outreach

- 2019 - 2021 **Space Science Journalist (Current in Space segment), *The Star Spot* podcast**
Curate, write and record stories about the latest research and news in astronomy. Episodes 177 - end of show.
- 2015 - 2018 **Python Workshop Lead, Sir Winston Churchill Secondary, Vancouver, B.C.**
Coordinated and delivered workshops to prepare students for national programming contest.

Media

- 2021 Jul **National Radio Astronomy Organization (NRAO) eNews**
Digital newsletter feature: “Molecular gas in high redshift galaxies”. https://science.nrao.edu/enews/14.7/index.shtml#molecular_gas
- 2020 Jul **SURP Student of the Week Feature**
Interview: <https://www.dunlap.utoronto.ca/surp-sow/sow12020/>

Technical Skills

JAX, Stan, Pytorch, SQL, R, C++, Rust, BLAS/LAPACK, L^AT_EX, regex, bash, git, Docker, Javascript, React